

HRIDIK HINGORANI

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Education

University of Illinois at Urbana-Champaign

Exp: May 2027

Grainger College of Engineering, Bachelor of Science in Computer Engineering, Minor in Mathematics

Relevant Coursework

ECE 470 (Introduction to Robotics), ECE 489 (Robot Dynamics and Control), ECE 486 (Control Systems), ECE 385 (FPGA Design), ECE 391 (Operating Systems)

Experience

ECE 385 Course Assistant

August 2025 – Present

University of Illinois at Urbana-Champaign

Urbana-Champaign, IL

- Supported 100+ students in SystemVerilog-based FPGA labs covering finite state machines, datapaths, memory systems, VGA output, simulation, synthesis, and hardware debugging
- Debugged student designs during office hours and lab sections, resolving issues in timing, reset logic, module interfaces, testbench behavior, and FPGA deployment

HealthArk Insights

May 2025 – August 2025

AI Software Development Intern

Mumbai, India

- Built Python-based document-processing features including summarization, web scraping, and dynamic formatting tools tailored to client specifications
- Integrated client feedback into production workflows by gathering requirements, validating features, and refining document-management functionality

Projects

3-DOF Robot Arm Obstacle Course Navigation | *MATLAB/Simulink, PID Control, Feedback Linearization* May 2026

- Constructed task-space PID controllers with hybrid force/position control on 3-DOF CRS arm for concurrent trajectory tracking and force regulation
- Generated MATLAB trajectory planners that navigated 6-step obstacle course with end-effector completion in under 12 seconds

UR3 Vision-Guided Manipulation and Image Drawing | *OpenCV, Python, Pose Estimation*

May 2026

- Implemented UR3 inverse kinematics using the product-of-exponentials formulation and integrated camera feedback for closed-loop visual guidance
- Built an OpenCV pipeline for pose estimation, object detection, and camera-to-robot calibration, enabling vision-based trajectory generation with sub-5mm drawing accuracy

BB-8 Inspired Spherical Robot with Inverted Pendulum Balancing | *CAD, MATLAB/Simulink, Embedded Control, Robotics*

- Designed and prototyped a spherical robot platform using inverted pendulum balancing principles for internal stabilization and locomotion control
- Developed CAD assemblies for the internal drive structure, actuator mounting, and sensor integration while evaluating 3D-printable component design
- Implemented reaction-wheel and closed-loop feedback control strategies using IMU sensor data for real-time stabilization and trajectory regulation

Reaction Wheel Inverted Pendulum Control | *MATLAB/Simulink, LQR, Observer Design*

December 2025

- Designed LQR controller with Luenberger observer for pendulum balancing, achieving settling to within 2% in under 500 ms with zero steady-state error
- Synthesized swing-up control using energy-based methods, achieving stable equilibrium with real-time DSP validation

Technical Skills

Control Systems & Robotics: pole placement design; feedback linearization for nonlinear robot dynamics; LQR, PD, PID, PI, hybrid force/position control, task-space control, MATLAB/Simulink

CAD & Mechanical Design: SolidWorks, Fusion 360, electromechanical prototyping, 3D printing, mechanical assemblies, tolerance-aware design

Embedded Systems & Real-Time OS: ARM Cortex-M microcontroller programming (C); RISC-V assembly, context switching, threading, condition variables, reader-writer locks, synchronization primitives, timer/alarm systems, device driver integration (VirtIO), FAT-based file systems and cache

FPGA & Hardware Design: SystemVerilog, Vivado, RTL simulation, testbenches, finite state machines, datapath design

Programming Languages: Python, C, MATLAB, SystemVerilog, RISC-V Assembly